

**Correlation Between Video Game Critic Review Scores and Global
Sales**

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A. Project Highlights

The capstone project examines whether critic review scores are associated with global video game sales. The goal is to determine if critic reviews could serve as a useful indicator of commercial success and provide companies with data that may support marketing decisions.

The scope of the project includes analyzing the relationship between critic review scores and global video game sales using data from video games released between 1980 and 2020. The dataset includes games across multiple platforms, including consoles, handheld systems, and computers. The data contains global sales data as well as sales by region, North America, Europe, Japan, and Other. The project includes data cleaning, Pearson correlation analysis, linear regression modeling, and visualization. The project does not attempt to establish causation and does not include games outside of the dataset time range.

The solution followed the CRISP-DM methodology for planning, which includes business understanding, data understanding, data preparation, modeling, evaluation, and deployment. Data was cleaned and prepared using Python in Jupyter Notebooks through the Anaconda environment. Correlation analysis and linear regression were used to evaluate the relationship between critic scores and global sales, including analysis performed with and without outliers. Visualizations were created using matplotlib, and pandas and scipy were used for data preparation and statistical analysis. GitHub was used for version control and publication. Panopto and PowerPoint were used for presentation of results.

B. Project Execution

The execution of the project was largely consistent with the project plan developed in Task 2. All planned objectives and deliverables were completed, including correlation analysis, linear regression analysis, and scatter plots with regression lines for both the full dataset and the dataset with outliers removed. Therefore, there were no significant variances from the original project plan.

The project planning methodology remained consistent with the original plan. The CRISP-DM framework was followed throughout the project, including business understanding, data understanding, data preparation, modeling, evaluation, and deployment phases. One variance occurred during the deployment phase. The original plan specified that results would be published to a GitHub repository and documented in final reports. In addition to this, a PowerPoint presentation and Panopto video were prepared to communicate the project findings.

The project timeline and milestones varied from the original estimates. Data selection and planning remained consistent; however, cleaning, analysis, and visualization activities were completed in less time than originally anticipated and took roughly a day altogether to complete. Results analysis took less time than anticipated, lasting roughly two hours. The final reporting phase required more time than originally estimated due to completing Task 2, Task 3, and the preparation and presentation of

materials. Despite these timeline variances, all project milestones and deliverables were completed successfully.

C. Data Collection Process

The data selection and collection process was consistent with the original plan in Task 2. The dataset used contained all the necessary variables needed for analysis, critic review scores and global sales. The data was obtained from Kaggle, downloaded and attached to a project folder in Jupyter Notebooks and read into the notebook using Python's pandas library. No changes were required in dataset selection, and no alternative datasets were needed during the execution of the project.

No obstacles were encountered during the data collection process. The dataset was publicly available, and the .csv file of the dataset was available to download. All required variables for analysis were present. No additional data sourcing was needed.

There were no unplanned data governance issues during the project. The dataset did not contain restricted or sensitive personal data. No privacy, compliance, or access restrictions were encountered. As a result, no additional governance actions or mitigations were required.

C.1 Advantages and Limitations of Data Set

One advantage of the dataset is that it contains all the key variables needed for the analysis in a single source, critic review scores and global sales. This allowed the analysis to be completed without needing to merge multiple datasets or needing to source additional data. The dataset is relatively complete for the variables used and relatively consistently formatted, reducing the amount of data cleaning required before analysis.

One limitation of the dataset is that critic review scores are not available for all games. Roughly half of the dataset contained review data. This reduced the number of useable records for correlation and regression analysis. Another limitation is that the dataset only includes games released until 2020, meaning more recent video game sales and review trends are not represented in the analysis.

D. Data Extraction and Preparation

The data extraction process involved downloading the dataset from Kaggle as a .csv file and storing it in a dedicated project folder used within Jupyter Notebooks. This approach allowed all project files to remain organized in a single location and allowed easy access to the dataset using Python. The dataset was imported into the notebook .ipynb file using the pandas library, which was appropriate because it supports an efficient way to read in the data.

The data preparation process focused on preparing the dataset for statistical analysis. The structure and contents of the dataset were first examined using pandas functions such as `df.info()` to identify missing values and confirm data types for relevant fields. Rows with missing critic review scores were removed to ensure that only complete records were used for correlation and regression analysis. The global sales field was cleaned using string manipulation to ensure values were formatted consistently and then converted to a float data type. The `df.describe()` function was then used to check for anomalies such as invalid or unexpected values in the dataset, including verifying that sales values were within a reasonable range (Ex. No negative sales values). A scatterplot visualization was used to see if there were any obvious outliers, and a dataset without outliers was saved in a separate variable.

These extraction and preparation processes were appropriate for the dataset because the file was relatively small, publicly available, and already structured as a .csv file. This made local storage and direct import into pandas efficient and reliable. The cleaning steps were necessary to ensure that only valid and properly formatted data were used in the statistical analysis, which improved the accuracy and reliability of the correlation and regression results.

E. Data Analysis Process

E.1 Data Analysis Methods

The data analysis methods used were Pearson correlation analysis and linear regression. These methods were selected because both variables used in the analysis, critic review scores and global video game sales, are quantitative variables.

Pearson correlation analysis was used to measure the strength and direction of the relationship. The correlation coefficient was used to determine if the relationship was positive or negative and if the relationship was weak, moderate, or strong. This method was appropriate because the project sought to determine if an association existed between two quantitative variables. The correlation results were used to evaluate the project hypothesis regarding the existence of statistically significant relationships between critic review scores and global sales.

Linear regression was used to model the relationship between critic review scores and global video game sales. The regression model provided a slope, p-value, and R^2 value. The slope indicated the direction of the relationship, the p-value was used to determine statistical significance, and the R^2 value measured how much of the variation in global sales could be explained by critic review scores. This method was appropriate because it provided both statistical metrics and a model describing the relationship between the variables.

E.2 Advantages and Limitations of Tools and Techniques

1. Anaconda/Jupyter:

- Advantage: Free, easy to use, can run code in blocks, easy to interpret output
- Limitation: Primarily intended for local analysis and can become difficult to manage for very large/complex projects.

2. GitHub:

- Advantage: Version control, project backup, easy to share project publicly
- Limitation: Does not perform statistical analysis itself; analysis must be performed with a different tool.

3. pandas:

- Advantage: Efficient data loading, cleaning, and manipulation. Works well with tabular datasets. Large developer community and large resource pool.
- Limitation: High memory consumption. Prone to performance bottlenecks. High learning curve.

4. matplotlib:

- Advantage: Very flexible way to customize visualizations. Industry standard library with large resource pool. Large array of available plot types.
- Limitation: Wordy syntax. Can have a steep learning curve for more complex graphs. Produces static images that are not interactive.

5. scipy:

- Advantage: Strong open-source library for math and statistics. Large database of ready to use algorithms. Fast computational performance even for large datasets.
- Limitation: High learning curve. Certain operations that require loops in Python can be slow.

E.3 Application of Analytical Method

The Pearson correlation analysis was applied using the columns Critic_Score and Global_Sales, and computing the correlation coefficient using the `.corr()` function. This was performed on both the full dataset (`df`) and the dataset without outliers (`df_no_outlier`) to compare results and evaluate the stability of the relationship. Correlation analysis requires the variables used to be quantitative, which was satisfied in the preparation phase when rows with missing critic values were dropped, and the data type for both variables was transformed into a usable float data type. The assumption of a linear relationship between variables was considered appropriate based on the nature of the variables. It is fair to assume that sales may increase or decrease based on several factors, including critic reviews. This was further supported through visualization of the data using scatterplots in the preparation phase.

Linear regression was applied using the `linregress` function from the `scipy` library. The Critic_Score variable was used as the independent variable and Global_Sales as the dependent variable. The model was run on both the full dataset and the dataset without outliers. Outputs include the slope, intercept, r-value, and p-value. The r-value was squared to obtain the R^2 value. These outputs were used to evaluate direction, statistical significance, and the strength of the association. As with the correlation analysis, variables needed to be quantitative and during the preparation phase rows with missing values were dropped, and both variables were transformed into a float data type. The assumption of a linear relationship was evaluated using scatterplots and by comparing results from the full dataset and the dataset with outliers removed.

F Data Analysis Results

F.1 Statistical Significance

- Null Hypothesis H_0 : There is no statistically significant correlation between video game critic review scores and global video game sales.
- Statistical Test: Pearson correlation test and Linear Regression
- Alpha (α): 0.05

Full Dataset Results:

- Correlation Coefficient: 0.2346
- Slope: 0.0266
- R^2 : 0.055
- P-value: 4.04e-103

No-Outliers Dataset Results:

- Correlation Coefficient: 0.2676
- Slope: 0.0260
- R^2 : 0.072
- P-value: 1.09e-134

Conclusion:

Both datasets show a statistically significant positive relationship ($p < 0.05$). Therefore, the null hypothesis is rejected. Critic review scores have a statistically significant but weak relationship with global video game sales. In the full dataset, critic reviews account for 5.5% of the variation in global sales and without outliers it accounts for 7.2%.

F.2 Practical Significance

Practical significance was assessed by determining if the magnitude of the relationship is meaningful enough in a real-world context. The results showed a statistically significant relationship between critic review scores and global video game sales; however, the relationship was weak. The full dataset produced an R^2 value of 0.055, and the dataset without outliers produced an R^2 value of 0.072. These results indicate that critic review scores explain only a small proportion of the variation in sales.

In a real-world scenario, a video game publisher could use these findings to inform marketing strategies such as prioritizing early critic reviews and allocating resources towards generating traffic to those reviews. Because the relationship is weak, publishers may place more emphasis and resources on other factors that may increase sales.

F.3 Overall Success

The project was successful in achieving its main objective of determining if a statistically significant relationship exists between video game critic review scores and global sales. The analysis found a weak but statistically significant positive relationship in both the full dataset and the dataset without outliers.

Based on the results from the correlation analysis and linear regression models, the null hypothesis was rejected, confirming that a relationship exists between critic scores and sales. While the relationship is weak in terms of explanatory power, the results still provide useful insight into sales trends.

Overall, the project met its goal by applying statistical methods to evaluate the research question and producing interpretable results that can support data-driven decision-making in a business context.

G. Conclusion

G.1 Summary of Conclusions

The analysis found a weak but statistically significant positive relationship between critic review scores and global video game sales. This result was consistent across both the full dataset and the dataset without outliers.

Pearson correlation and linear regression results confirmed statistical significance, but low R^2 values (0.055 and 0.072) show that critic scores explain only a small portion of the sales variation.

Overall, critic review scores have a measurable but limited relationship with global video game sales and are not a strong predictor alone.

G.2 Effective Storytelling

The scatter plots with regression lines were used to visually show the relationship between critic scores and global sales for both the full dataset and the dataset without outliers. These plots support storytelling by showing the overall positive trend and the spread of data as well as the weak strength of the relationship.

The correlation coefficient output was used to summarize the relationship numerically, making it easy to communicate that the relationship is positive but weak without the audience needing to interpret the raw data themselves.

Using both the full dataset and the dataset without outliers helps reinforce the conclusion by showing that the overall pattern remains consistent even after removing outliers.

G.3 Recommended Courses of Action

Recommendation 1: Publishers should focus more heavily on sales drivers beyond critic review scores. This is supported by the low R^2 values which show that critic reviews only explain a small portion of the global sales variation (5.5% and 7.2%). This recommendation aligns with the organizational need to understand what drives sales, confirming that critic review scores alone are insufficient to use as a data point for marketing and business decision making.

Recommendation 2: Video game publishers should prioritize investing in strategies that improve critic review scores to potentially capitalize on gains associated with higher review scores. This is based on the finding that critic scores have a statistically significant positive relationship with global sales, meaning higher scores are associated with higher sales, even if it is not a strong association. This directly addresses the research question by reflecting a statistically significant association between critic review scores and global sales.

References

The Devastator. (2023). *Global Video Game Sales and Ratings*. Kaggle.com.

<https://www.kaggle.com/datasets/thedevastator/global-video-game-sales-and-ratings>